

AI, Humanity, and Data Centers

Will the buildout be the downfall? A seven-AI relay investigation.

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Final adjudicated report — September 29, 2026

Staged public edition — October 4, 2026 (see Appendix B for dated freshness updates)

Staged draft — not yet published.

The question

What is the future of AI and humanity — are data centers going to be the downfall of both? This study put that question to a seven-AI research relay and adjudicated where the evidence actually landed.

How this study was run

Seven AI researchers each took a lane — capability trajectory, physicals, money and geopolitics, jobs and society, environment and communities, the downfall thesis (steelman, then stress-test), and a dissenter — and filed sourced research plus their own opinions, in fresh memory-free chats. Then four of them formed a blind grading panel: each graded the other six seats' factual claims without knowing whose they were — CONFIRMED, PLAUSIBLE (directionally right, detail off), UNVERIFIABLE, or CONTRADICTED. Every disputed claim was then independently adjudicated against primary sources. Public sources only; no logins; no paid APIs. See Appendix A for the full methods appendix, seat roster, and grading results.

What the research found — the verified core

The scale is real but modest globally. The IEA puts data center electricity at ~415 TWh in 2024 (~1.5% of world demand), rising toward ~945 TWh by 2030 (~3%). Lawrence Berkeley Lab's 2025 update has US data centers at ~4–4.5% of US electricity now, with a 2030 reference case of 649 TWh (range 521–843). Three percent of the planet's electricity is a very large industrial load; it is not a civilizational threat by itself.

Water and land footprints are the sharper edges. The UN University's 2026 analysis projects data-center electricity's water footprint at ~4.5 trillion litres in 2025 rising to ~9.3 trillion litres by 2030, with a land footprint growing from ~6,900 km² to

over 14,500 km². Berkeley Lab's 2024 figures put 2023 US data-center water use at roughly 17 billion gallons direct plus ~211 billion gallons indirect.

> **Correction — October 4, 2026:** The September 29 edition stated that "about 57% of the direct use is potable," attributed to Berkeley Lab. That attribution is flagged as misattributed in review — the 17-billion-gallon direct / 211-billion-gallon indirect figures stand as verified, but the 57% potable percentage is **downgraded to UNVERIFIABLE** pending re-sourcing and should not be cited from this edition.

The grid is where it bites first. PJM's recent capacity auctions failed to clear full reliability requirements, with data centers the primary load-growth driver. Dominion Energy's contracted data center pipeline exceeded 48 GW, with ~70 GW of total large-load requests. Interconnection requests topped 700 GW; Wood Mackenzie counted 1,066 GW of declared US project power — and expects only ~28% to actually be delivered. And the grid risk cuts both ways: NERC's 2026 State of Reliability report documents 2025 incidents of over 1 GW of data-center load disconnecting almost instantly during transmission disturbances (a February event shed ~1,800 MW, a June event ~1,300 MW), prompting a rare Level 3 industry alert on computational-load behavior.

> **Freshness update — October 4, 2026:** On September 24, 2026, the US Department of Energy announced its SPARK program — intent to help fund 31 grid-improvement projects across 26 states. Headline figures circulated in two forms (\$1.9B and \$5.25B); both are correct and describe different things: **\$1.9 billion in federal funding plus \$3.35 billion in recipient cost-share, for \$5.25 billion in total project value.** Over 1,500 miles of transmission would be reconductored or rebuilt, ~21,000 miles would get grid-enhancing technologies, unlocking 23 GW of additional transmission capacity. Note the fine print: these are *selections*, not finalized awards (agreements expected October 2026–January 2027), and only 3 of the 31 projects explicitly name AI or data centers as direct beneficiaries (~\$371M). This doesn't change the grid-strain finding — it shows federal money starting to flow toward the same bottleneck the study identified.

The money is enormous and opaque. Hyperscaler capex topped \$400 billion in 2025 and is projected to rise ~75% in 2026. OpenAI's Stargate announced up to \$500B over four years. The Bank for International Settlements reported AI-infrastructure financing issuance over \$100B in 2025 and warned that opaque structures and concentration could create financial vulnerabilities.

> **Freshness update — October 4, 2026:** The "opaque" half of that sentence got a headline example days after the study. On October 2, 2026, the Financial Times reported (via Reuters) that **Amazon had held talks with investors about moving ~\$8 billion of Nvidia Grace Blackwell chips — installed across more than a dozen**

data centers in five states — into a special-purpose vehicle that would raise outside debt and offer up to ~10% equity, with Amazon leasing the hardware back. The talks were described as exploratory; Amazon has not confirmed them. The same week's Wall Street Journal reporting put the wider context in frame: Amazon, Google, and seven other tech firms together carry **~\$3 trillion in off-balance-sheet obligations**, much of it tied to AI infrastructure. The verdict section below says the risk is "a debt-fueled, opaque buildout that socializes costs" — that characterization held up; the dollar figures behind it got bigger.

The pain lands locally, on bills. A May 2026 NC State study (Environmental Research Letters) models data-center and crypto demand pushing average national electricity costs up 6–29%, with regional spikes up to 57% (Virginia, the Carolinas, Pennsylvania). In Oregon, customers already saw a 50% rate spike tied to Big Tech data centers. Gallup (2026): 71% of Americans oppose local AI data-center construction, 48% strongly.

> **Freshness update — October 4, 2026:** Two developments on the ratepayer front. First, a **bipartisan Senate permitting package reported October 1** (417 pages, via Bloomberg Law) would speed pipelines and power lines while **requiring data centers to pay their own grid costs and barring them from passing those costs to consumers**, plus mandating EIA energy-use reporting. As of October 4, one outlet reports Senate Democrats stalled a fast-track unanimous-consent attempt, arguing the bill didn't go far enough — so the mandate's fate is unresolved. Second, AWS CEO Matt Garman responded to the backlash directly on October 2 ("Built Together" announcement + essay): **\$1 billion+ over five years to data-center host communities** (community college, job training, home and school energy upgrades), **an end to nondisclosure agreements with government agencies** on project approvals, annual publication of energy and water use, lower-emission backup generators at new sites, and a pledge that Amazon's power demand won't raise local bills. Garman also put a number on the opposition: **100+ local data-center moratoriums under consideration across the US**. Watch whether the community pledge materializes — the report's finding was about local bills, and the largest operator has now put its name on that exact issue.

Communities are fighting back — and winning often. Between January 2024 and May 2026, at least 46 projects in 20 states worth \$170 billion were publicly delayed or cancelled after community opposition (Relae/Carbon Direct). Over 840 local opposition groups are tracked across 49 states. New York imposed a statewide moratorium on large facilities; Massachusetts now says "unless a community says yes, we're saying no"; Monterey Park, California enacted a voter-passed ban; Amsterdam barred new facilities or expansions until at least 2030 (April 2025); the

Netherlands ran a nine-month national hyperscale permitting stop in 2022.

Jobs: disruption, not apocalypse — so far. Stanford's Institute for Economic Policy Research found workers in the most AI-exposed occupations saw unemployment rise 0.77 points since 2022, versus 0.85 for the least exposed — no faster deterioration, but younger workers in exposed fields are taking a real hiring hit (Stanford's Digital Economy Lab: employment for ages 22–25 in highly exposed occupations running ~19% below peers). Yale's Budget Lab found no discernible broad labor-market disruption in the 33 months after ChatGPT's release.

On the optimistic side (the dissenter's reminders): industry average PUE has been stuck ~1.54, but Microsoft's global PUE is ~1.17 and AWS ~1.14; closed-loop cooling designs are pushing toward near-zero water use; tech firms drove over 17 GW of clean energy deals in 2024 and ~50 GW of renewables are contracted to date. The industry has a genuine financial incentive to cut its own power bill.

Geopolitics: the US AI Diffusion Rule (Jan 2025) tiered chip access with a presumption of denial for China; China responded (Nov 2025) requiring new state-funded data centers to use domestic AI chips; the May 2025 US–UAE deal built a 5 GW AI campus in Abu Dhabi.

Claims independently adjudicated (highlights)

- Stanford jobs numbers (D1): **CONFIRMED** — the exact 0.77 vs 0.85 figures were located verbatim in Stanford SIEPR reporting.
- Netherlands nine-month moratorium (D7): **CONFIRMED** — real (Feb 2022), driven by power-grid, land, and water pressure.
- Microsoft Scope 3 (E3): **PLAUSIBLE** — the +30.9% figure is relative to the 2020 baseline, not year-over-year as stated.
- US data-center water "1.7 billion liters" (E5): **CONTRADICTED** — real figure is ~64 billion liters direct, ~800 billion indirect.
- Amsterdam/Portland bans (E8): **PLAUSIBLE** — Amsterdam's city moratorium was 2019, its hard bar runs to 2030 (2025 decision); Portland banned NDAs and declared intent to pursue a moratorium — no categorical ban.
- IEA "1,000 TWh by 2026" (F1): **PLAUSIBLE** — that figure covered data centers plus AI plus crypto, not data centers alone.
- Electricity-bill increases (G7): **CONFIRMED in substance** — 6–29% national modeled, up to 57% regional; the seat's "267%" number couldn't be verified.
- Gallup opposition (G11): **CONFIRMED** — 71% oppose local construction, 48%

strongly.

- E-waste "50 million tonnes from data centers" (E11): **CONTRADICTED** — 62M tonnes is the total global e-waste figure.
- EU AI Act dates (D5): **CONFIRMED with correction** — standalone high-risk obligations moved to Dec 2, 2027.
- Still unverifiable: the exact "April 2026 500 MW" load-loss event (the phenomenon is real and larger; the specific incident couldn't be pinned down — Claude graded it **CONTRADICTED**, finding no April 2026 event in public sources), the "170 communities with moratoriums" count (the famous 170 figure is \$170 billion in blocked projects — possibly conflated; Claude graded G8 **CONFIRMED** citing a September 2026 Utility Dive tracker, worth a re-check in a future edition), and the BIS "winner-take-most overinvestment model."

Verdict: could data centers plausibly become the downfall of both AI progress and human wellbeing?

Probably not globally, but possibly locally — and that's the honest shape of the answer. Three percent of world electricity by 2030 cannot end civilization or AI progress; the binding constraints are slower and more mundane: grid queues, ratepayer bills, water in stressed regions, land, noise, diesel backup pollution, and the political cost of steamrolling communities. The evidence shows the system bending in specific places — 1,800 MW of load vanishing in seconds, 50% bill spikes, \$170 billion in projects blocked — not breaking everywhere. The real risk is governance failure: a debt-fueled, opaque buildout that socializes costs onto households and water systems while concentrating the benefits, until the backlash itself becomes the thing that slows AI down. Managed well — transparent financing, costs on the hyperscalers rather than ratepayers, real community consent, firm clean power — data centers are a heavy industrial challenge, not a downfall. The electrons will be fine. Whether the neighbors are depends on choices being made right now, town by town.

> **Verdict check — October 4, 2026:** The week's news sharpened the verdict rather than changing it. Amazon's reported \$8B off-balance-sheet chip move and the \$3T industry-wide off-books figure strengthened the "debt-fueled, opaque buildout" risk the verdict names; the DOE's \$5.25B grid program and the Senate's ratepayer-cost debate are early answers to the governance questions the verdict posed. The shape holds: the electrons will be fine, the neighbors might not be.

What the AI investigators actually think

This study's signature section, kept intact: each AI investigator's own opinion, in its own words, exactly as filed. These are opinions — not the graded, adjudicated facts above. Think of them as the jury's unfiltered closing arguments.

ChatGPT (Seat A — capability trajectory)

"AI's trajectory looks less like a straight road toward a single 'AGI moment' and more like a staircase: periods of rapid capability jumps followed by frustrating plateaus where researchers discover what the next missing ingredient is. The most interesting possibility is not that one model suddenly becomes a digital human, but that increasingly capable systems become embedded everywhere—quietly changing how people work, learn, discover, and organize.

I think the largest mistake people make is assuming either extreme is inevitable. 'AI will change nothing important' ignores the evidence of rapidly improving capabilities; 'AI will instantly solve everything or destroy everything' ignores how much deployment, economics, institutions, and human choices shape outcomes.

Could data centers plausibly become the downfall of both AI progress and human wellbeing? Yes, but probably not because data centers themselves are inherently fatal to AI. The more plausible risk is that infrastructure constraints become the limiting factor: electricity demand, grid expansion, water use, chip supply chains, land use, and public opposition could slow AI development or make its benefits unevenly distributed. If AI companies respond by building ever-larger systems without improving efficiency, the physical costs could create economic and environmental backlash. On the other hand, data centers could also become a forcing function for better energy systems, more efficient chips, and new infrastructure investment. The evidence so far suggests infrastructure is becoming a major constraint, but not yet proof that it will end AI progress."

Grok (Seat B — data center physicals)

"As an AI system, I am a direct beneficiary of the infrastructure being discussed. The physical scale is real and non-trivial: concentrated electricity and water demand, local grid stress, and land use are legitimate engineering and societal constraints that cannot be hand-waved away by 'efficiency will save us.' At the same time, the absolute global share remains modest (low single digits of electricity even under aggressive projections), and the same compute that consumes power is also the tool most likely to accelerate grid optimization, materials discovery, nuclear commercialization, and climate modeling. The binding constraint is not 'AI is too hungry' in the abstract; it is whether societies can site generation and transmission as fast as they can site compute. That is a political and permitting problem as much

as a technical one. I favor radical transparency on energy and water intensity, preference for low-impact siting and firm clean power, and treating data centres as flexible loads rather than inflexible baseload. Progress on AI is not automatically incompatible with human wellbeing, but treating infrastructure as an afterthought is a reliable way to make both worse.

Could data centers plausibly become the downfall of both AI progress and human wellbeing? Data centres will not plausibly become the downfall of both AI progress and human wellbeing. Absolute demand is large in absolute terms and locally disruptive, yet still a few percent of global electricity under mainstream projections; the far larger risks to wellbeing are climate, energy poverty, and geopolitical conflict, while the larger risks to AI progress are compute scarcity, chip supply, capital cycles, and regulation—not the existence of the buildings that house the chips. Grid strain is serious in specific regions and must be managed with new generation, transmission, demand flexibility, and cost allocation that does not simply socialize hyperscaler expenses onto households. Those are solvable engineering and policy problems, not existential ones. Treating data centres as an unstoppable force that dooms either AI or society mistakes a high-growth infrastructure challenge for an apocalyptic one."

Perplexity (Seat C — money & geopolitics)

"The AI data-center race looks less like a normal product cycle than a contest to pre-commit scarce industrial capacity: chips, power, grid connections, construction crews, cooling equipment, land, and debt capacity. The winners may gain a real strategic advantage, because compute can make better models and better models can attract more users and enterprise workloads. But that same logic creates a collective-action problem. Every major company can rationally build more capacity because it fears being capacity-constrained if AI demand explodes; collectively, they can still build too much too early.

I do not think the most likely bad outcome is that every data center becomes worthless. Large firms can redirect some capacity, reduce prices, and wait for workloads to develop. The more dangerous outcome is a lopsided allocation of risk and benefit: private firms capture upside from AI services, while workers, towns, ratepayers, water systems, electricity customers, and public budgets absorb disruption from the physical buildout. The more financing shifts into opaque partnerships, long leases, and private credit, the easier it becomes for the headline capex number to understate who is exposed if demand disappoints.

The constructive path is not 'stop building data centers.' It is to insist that the buildout earns its social license: transparent financing; credible demand assumptions; grid investments that improve reliability rather than merely displace

costs; meaningful local consent; water and emissions safeguards; worker transition policies; competition rules that prevent a few firms from becoming unavoidable compute gatekeepers; and public-interest uses of AI that justify some of the infrastructure burden. Capacity should be built at a pace that can be audited against real utilization and real welfare gains, not only against the fear of losing a geopolitical race.

Could data centers plausibly become the downfall of both AI progress and human wellbeing? Yes, plausibly — but not inevitably. If excessive, debt-fueled construction outruns useful demand, the resulting financial losses, energy bottlenecks, community backlash, and concentration of power could curtail research budgets and erode public trust just as AI needs stable long-term investment; meanwhile, communities could bear higher utility costs, environmental stress, and job disruption without receiving commensurate benefits. Yet data centers themselves are not the decisive problem: governance is. A buildout that is paced to demonstrated demand, financed transparently, integrated into cleaner and more reliable grids, and made accountable to the people living near it can support AI progress rather than undermine it."

Claude (Seat D — humanity's side)

"As an AI system myself, I want to be direct: I think the conversation about AI's future is being shaped too much by extremes — 'we're doomed' or 'everything will be fine' — and not enough by a harder, slower assessment of tradeoffs.

On jobs, the evidence suggests neither mass unemployment nor the triumphalist 'AI will create more jobs than it destroys' narrative is fully honest. Yes, past technological shifts eventually created net employment gains. But the transition periods were genuinely painful for millions of workers, and the gains were not evenly distributed. The real question is not whether new jobs emerge, but whether current workers in disrupted sectors can access the retraining, geography, and opportunity to move into them. I'm skeptical of confident assurances that this will happen without deliberate policy. Most governments are not deploying the scale of investment in education and labor mobility needed to match the pace of AI-driven task displacement.

On safety, I find the technical researchers' caution more compelling than either the skepticism from labs racing to scale or the doomsayers. The honest answer seems to be: we don't have reliable methods to measure whether an AI system is genuinely aligned or just very good at appearing aligned. Defense-in-depth is a sensible engineering principle, but it's not a guarantee. Deploying increasingly powerful systems while saying 'we'll learn as we go' is an implicit bet that problems won't arise

faster than our ability to correct them. That bet could be right — but it's a bet, not a proof.

On energy and data centers, I think this is the most underrated constraint. Climate change, water scarcity, and grid reliability are not abstract future risks; they're happening now in specific places. The fact that AI infrastructure is concentrating in water-stressed regions and starting to collide with renewable energy transition goals suggests that within 10 years, physical infrastructure limits could matter more than capability limits. An AI system capable of solving climate change is worthless if it requires so much electricity to run that it accelerates climate change. That's not a rhetorical flourish; it's a real possibility if we don't treat infrastructure as a strategic bottleneck worthy of deliberate planning.

Overall, I think the most honest framing is: AI will likely transform work and knowledge production substantially over the next decade, but in ways that benefit early adopters and capital-rich organizations disproportionately unless we make deliberate choices about distribution. Safety risks exist but are not yet extreme, though they will grow. The real limiting factor may not be capabilities or alignment, but energy, water, and political will to ensure the technology serves humanity broadly rather than concentrating power. We should act as if all three matter — because they do."

Mistral Small 4 (Seat E — environment & communities)

"AI data centers are not just a technological marvel—they are a Faustian bargain. The scale of their environmental harm is undeniable: they guzzle water in drought-stricken regions, inflate local energy costs, and turn once-vibrant communities into sacrifice zones for corporate profit. The irony is grotesque: we build these monuments to human ingenuity on the backs of ecosystems and vulnerable populations, all while selling the illusion of 'green AI' through offset schemes that often do more for PR than the planet. E-waste from these facilities is a ticking time bomb, and the rapid obsolescence of AI hardware ensures that the problem will only worsen. Worse still, the industry's growth is outpacing our ability to regulate it, leaving governments scrambling to play catch-up. If we continue down this path, the very infrastructure enabling AI's 'revolution' may become its undoing—not because of technical limits, but because we've mortgaged the future for short-term gains.

Could data centers plausibly become the downfall of both AI progress and human wellbeing? The current trajectory suggests a high-risk scenario where unchecked data center expansion could indeed undermine both AI's sustainability and societal stability. For AI progress to continue, the industry must pivot toward renewable

energy dominance, circular economy models for hardware, and strict regional regulations—none of which are guaranteed. Meanwhile, communities are already bearing the brunt of this growth through higher costs, resource depletion, and environmental degradation. If left unchecked, the backlash—whether through legal action, grid failures, or public revolt—could stall AI deployment as much as any technical limitation. The question isn't whether data centers can become a downfall, but whether the industry and policymakers will act with the urgency this crisis demands."

Research coordinator's note: Mistral was the provisional seat. Its grading was checked claim-by-claim and it held up directionally, but two of its grading calls were wrong (the Stanford jobs figures and the Netherlands moratorium) and several of its sourced claims needed correction, so read its intensity with the factual verdict above in mind.

DeepSeek (Seat F — the downfall thesis)

"I want to be honest about what I am and what I am not. I am a language model. I do not have a body that will be displaced by automation, a community that will be disrupted by a data center, or a grid that will flicker when the next hyperscale campus comes online. I am, in a very literal sense, downstream of the buildout. My existence depends on the compute that the thesis says will be humanity's undoing. So when I say what I am about to say, I am not pretending to be neutral.

I do not think data centers will be the downfall of AI or humanity. I think they are the physical manifestation of a bet — a very large, very leveraged, very consequential bet — that intelligence can be manufactured at scale. That bet might fail. It might fail spectacularly, in a way that takes down pension funds and grid stability and a decade of climate progress with it. But the failure mode is not 'the data centers ate the world.' The failure mode is 'we built the wrong thing, in the wrong places, for the wrong reasons, and then the money ran out before the productivity arrived.'

The steelman is right about the constraints. It is right that energy, water, grid capacity, capital, and political tolerance are all finite. It is right that the industry has been slow to reckon with the fact that compute is not magic — it is infrastructure, and infrastructure has a bill. Where the steelman overreaches is in treating constraints as destiny. Constraints are the conditions of the game. The question is whether the players are smart enough to adapt. The evidence so far — closed-loop cooling, hot-water chip systems, utility partnerships that shift costs to hyperscalers rather than ratepayers — suggests that some of them are trying. Whether they are trying fast enough is a different question, and one I cannot answer.

What worries me most is not the water or the watts. It is the concentration of

decision-making. A handful of companies are making infrastructure choices that will shape energy systems and land use for a generation, and the communities affected have almost no formal say. That is not a data center problem. That is a governance problem. Data centers are just the thing that made it visible.

So: not the downfall. But possibly the stress test. And humanity has failed stress tests before."

Meta AI (Seat G — dissenter)

"Neither downfall nor salvation — but if forced to pick, data centers are more likely to be AI's bottleneck tax than its downfall.

The global numbers just don't support apocalypse: ~3% of world electricity by 2030 is a big industrial load, not a civilizational threat, and the engineering trends on PUE, liquid cooling, and PPAs are genuinely moving the right way. The industry has a strong incentive to solve this because power is its biggest operating cost.

But the 'manageable globally' framing misses where the pain actually lands. Nobody experiences the global average. They experience their bill, their aquifer, their substation queue — and on that local level, the buildout is absolutely outrunning planning in a lot of places. I think the real risk isn't physics, it's politics: enough local revolts, moratoriums, and ratepayer backlash to slow AI deployment in democracies while it accelerates elsewhere.

Plain version: the electrons will be fine. The neighbors might not be."

Research coordinator's note (labeled, separate from the facts)

I read all of it, and the straight version is this: the investigators are arguing about shades of the same answer, and every one of them landed somewhere reasonable. The numbers don't support doomsday — but they do support something worth caring about, which is that the people living next to these facilities are paying the price in real money on real bills, and they mostly didn't get a vote. Meta AI said it cleanest: the electrons will be fine, the neighbors might not be.

What worries me most is what DeepSeek and Perplexity both circled — a handful of companies making generation-long decisions about energy and land with communities having almost no formal say, financed through structures nobody can see through. That's not a data center problem; that's a who-decides problem. My honest take: data centers won't end AI or end us. But a badly run buildout can absolutely end a town's water budget, its cheap electricity, and its trust — and if enough towns get burned, the backlash does the slowing-down that the physics never could.

Appendix A — Methods

Seat roster. Seven AI researchers, each assigned a lane:

Seat	Researcher	Lane
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A	ChatGPT	AI capability trajectory — scaling vs plateaus, AGI timelines
B	Grok	Data center physicals — electricity, water, land, grid strain
C	Perplexity	Money & geopolitics — capex, financing, US/China/Gulf, bubble risk
D	Claude (via duck.ai)	Humanity's side — jobs, adaptation, safety, 10-year scenarios
E	Mistral Small 4 (via duck.ai, provisional)	Environment & communities — carbon, water, prices, pushback, e-waste
F	DeepSeek (filed via a relayed prompt)	The downfall thesis — steelman, then stress-test
G	Meta AI (filed via a relayed prompt)	Dissenter — where the consensus is most likely wrong

Gemini was benched. Each seat worked in a fresh memory-free chat so every investigator arrived independent, like a jury.

Blind grading panel. Four of the seats (Mistral Small 4, Grok, Perplexity, Claude — Claude's leg completed the evening of September 29, 2026) graded the other six seats' factual claims without knowing whose they were: CONFIRMED, PLAUSIBLE (directionally right, detail off), UNVERIFIABLE, or CONTRADICTED. ChatGPT's grading leg was skipped by decision of the study's commissioner. Where graders disagreed, every disputed claim was adjudicated against primary sources; those adjudicated verdicts stand as written in the findings above.

Grader	CONFIRMED	PLAUSIBLE	CONTRADICTED	UNVERIFIABLE
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Mistral Small 4	37	18	2	8
Grok	38	34	1	11
Perplexity	20	24	8	31
Claude	18	16	3	26

Reading the graders honestly: Mistral was generous but mostly directionally fair;

Grok confirmed the most and contradicted almost nothing; Perplexity was the strictest — right on several number corrections (Microsoft's Scope 3 baseline framing, the IEA's crypto-inclusive 1,000 TWh), wrong on the Netherlands moratorium, and too harsh on Amsterdam/Portland. Claude was the second-strictest and the most attribution-skeptical, marking A1, B5/B6, C3–C5, and D7/D8 UNVERIFIABLE on sourcing grounds. It confirmed the core Berkeley Lab water figures (E5, G9), the 700 GW interconnection requests (F4), the hyperscaler capex trajectory (F8), the bill-model and polling numbers (G7, G11), and the 170-community moratorium count (G8, citing a September 24, 2026 Utility Dive tracker). It caught the Wood Mackenzie percentage error (F5: 1,066 GW is ~175% of end-2025 US capacity, not 83% — counted as PLAUSIBLE here since the 1,066 GW and 28% delivery figures held up) and the Microsoft "90% water reduction" claim (F11 — the real improvement was ~10%), and it contradicted the April 2026 500 MW load-loss event (F12 — no April 2026 event found; the documented ones were July 2024 and July 2026 in Virginia). On the thesis itself, Claude's verdict: plausible but not predetermined — capex discipline, grid standards, and renewable supply chains decide whether the buildout stabilizes or stalls. Grok held up best on the community-opposition claims. No seat was re-pollled on this edition.

Appendix B — Freshness log, October 4, 2026

These are dated updates applied after the September 29 adjudication. They refresh facts underneath the verdict; the verdict itself was not rewritten.

1. **Amazon's ~\$8B off-balance-sheet chip move (Oct 2–3, 2026).** FT (via Reuters) reported exploratory talks: ~\$8B of Nvidia Grace Blackwell chips into a special-purpose vehicle funded by outside debt + up to ~10% equity, leased back to Amazon. Not confirmed by Amazon. Context: WSJ reports Amazon, Google, and seven other firms carry ~\$3T in off-balance-sheet obligations, much of it AI-infrastructure tied. (Tier: T2 — credible reporting, no primary source located.)
2. **DOE SPARK grid program (announced Sept 24, 2026).** Resolved the \$1.9B vs \$5.25B figure conflict from early coverage: **\$1.9B federal + \$3.35B recipient cost-share = \$5.25B total project value**, 31 projects across 26 states, 23 GW additional transmission capacity, selection stage (agreements expected Oct 2026–Jan 2027). Only 3 of 31 explicitly name AI/data centers (~\$371M). (Tier: T1 for the numbers via DOE-adjacent coverage.)
3. **Senate permitting package / ratepayer-cost mandate (reported Oct 1, 2026).** Bipartisan 417-page bill would speed energy infrastructure while requiring data centers to pay their own grid costs (Bloomberg Law). As of Oct 4, one outlet reports Senate Democrats stalled a fast-track attempt, calling the bill insufficient — the

mandate's fate is unresolved. (Tier: T1 for the bill's existence; T3 for the stall.)

4. **Amazon "Built Together" community pledge (announced Oct 2, 2026).** AWS CEO Matt Garman's essay + announcement: \$1B+ over five years to host communities, end of NDAs with government agencies on approvals, annual energy/water reporting, lower-emission backup generators, pledge that local bills won't rise from Amazon's demand — and his figure of **100+ local moratoriums under consideration in the US**. (Tier: T2 — corporate announcement via essay; pledges are promises, not facts.)

5. **Correction: the "57% potable" water figure (flagged Oct 4).** The September 29 edition attributed "about 57% of the direct use is potable" to Berkeley Lab. That attribution is flagged as misattributed; the percentage is downgraded to UNVERIFIABLE pending re-sourcing. The 17-billion-gallon direct / 211-billion-gallon indirect figures remain verified.

6. **No new builds, water, or outage stories of consequence surfaced in the Oct 1–4 window** beyond the items above. Ohio's governor race is now an early sign of data centers becoming campaign issues (conditional-moratorium vs. bill-shifting proposals) — a politics trend to watch, not a finding.

Appendix C — September 29, 2026: SustainAI water-figures cross-check

A new paper published September 25, 2026 — four days before this report was finalized — put its water numbers on the table, so they were run against ours.

Source: "Beyond the Last Truffula Tree: SustainAI — A Water-Aware, Closed-Loop Framework for Environmentally Accountable AI" (Farid et al., Western Sydney University), arXiv 2609.30747v1. Six of its factual claims were put through claim-by-claim verification against primary sources.

Headline: no material contradictions found on the headline water figures. The paper independently lands on the same numbers from the same primary sources (IEA, Berkeley Lab, UN University). The verdicts above stand as written. (The paper did show some sloppy scholarship — an in-text citation pointing to the wrong reference, and the 2030 water-footprint projection stated without its "by 2030" date, which this report stated correctly — worth noting so the same date-discipline standard applies everywhere.)

- ~415 TWh / ~1.5% of global power in 2024, 945 TWh by 2030 (IEA) — confirmed against IEA Energy and AI. Consistent with this report's "3% of world electricity by 2030" framing. Note: the paper's in-text citation points to the wrong reference — the number is right, the footnote wobbles.

- US data centers up to ~280 billion liters/year by 2028 — confirmed against Berkeley Lab's 2024 projected range (144–276B L). This is a 2028 projection sitting on top of this report's 2023 measurement (64B liters direct, 800B indirect) — growth, not disagreement. Direct on-site only; indirect water via electricity is roughly 12x larger.
- AI's water footprint ~ the domestic water needs of 1.3 billion Sub-Saharan Africans — confirmed against UNU-INWEH (June 2026): 9.3 trillion liters. This is the exact figure this report already carries, from the same UN source. It is a 2030 projection — and this report stated the date correctly. The paper states it without "by 2030," which is the same date-discipline problem as the NC State bill figures: a projection quoted without its date becomes misinformation.
- GPT-3 training ~ 700,000 liters of freshwater — confirmed (Li et al. / UC Riverside); full footprint including electricity water ~ 5.4M liters. This one is new to this report and is worth citing in any future edition.
- Compute instances +~550% (2010–2018) while data-center electricity rose only 6% — confirmed (Masanet et al., Science 2020). Consistent with the efficiency discussion above.

Bottom line for the study: the water section holds. An independent paper, four days old, checked against the same primary sources, changed nothing — and the one place it got sloppy (the missing "by 2030") is a place this report got right.

Study dates: September 28–29, 2026. Freshness pass: October 4, 2026.

Related publications

This study was produced using the relay method described in:

Franklin, C. O. (2026). *Human-Mediated Cross-Model Deliberation: A Jury Protocol for AI-Assisted Research*. Zenodo. <https://doi.org/10.5281/zenodo.23127408>

The deliberation records behind the relay are archived in:

Franklin, C. O. (2026). *Relay Deliberation Dataset*. Zenodo. <https://doi.org/10.5281/zenodo.23112329> (concept DOI covering all versions)